

TIER 0

SIMULATOR



YOU FLY

FPV sim rigs in the IC · Liftoff · Micro Drones · SkyDive · FPV Labs · PicaSim

TO GET HERE

Have a name

ON THIS SHEET

7 badges · all of them to pass Gate 1

2 QUIZ · 2 BENCH · 2 WITNESSED · 1 AUTOMATIC

GO

START

Add your name at the kiosk

TIER 0 · PT. 1

Simulator Flight

7.5 hours logged in the simulator.

AUTOMATIC

Earned automatically by the kiosk at 7.5 h across the five flight sims. Ground School and bench time do not count.

WHY Acro muscle memory where crashes cost nothing.

TIER 0 · PT. 3

Safe Flight

Know and follow the Brophy FPV safe-flight rules, and the basics of the airspace over Brophy.

QUIZ

5 LESSONS, THEN THE QUIZ

1. Who's flying, who's watching
2. The Brophy FPV rules sheet
3. Class B, 400 ft and Sky Harbor
4. B4UFLY, LAANC and TRUST
5. Go / No-Go **DRILL**

10 questions · 80 % passes · no sign-off needed · retake any time

WHY The baseline code of conduct for real hardware, and why the campus sits under Sky Harbor's Class B.

TIER 0 · PT. 5

Building

Fully build a basic drone from parts.

BENCH

AT THE BENCH · 6 STEPS

Parts checklist › Frame assembled › Motors on › Stack mounted › Soldered › Smoke-stopper first power-up

Photo DM'd to mtucker27@ · a mentor signs it off

WHY Members can fix what they break.

TIER 0 · PT. 7

Line of Sight

Without goggles: take off, hover and land a whoop in angle mode. Level flying only, supervised.

WITNESSED

THE STANDARD

A mentor stands beside you: a controlled take-off to head height, a 20-second hover inside a two-metre box, a slow nose-in turn, and a landing on the pad.

Practise first in Ground School: Which way is left? · Gate Run · Lost orientation recovery

WHY If video cuts out, the pilot can still level out and land.

1

2

TIER 0 · PT. 2

Proficient Flight

In the sim: orbit, Split-S, gaps, and finish a race.

WITNESSED

THE STANDARD

A mentor watches you in Liftoff: a clean orbit around a gate, a Split-S, three gaps without a crash, and a finished race.

Practise first in Ground School: Four channels, two sticks · Rate mode against angle mode · Hover Trainer · Throttle discipline

WHY Real control, not just hovering. Sim-only, so it can gate the first real flight.

TIER 0 · PT. 4

Battery Care

Balance-charge, store at storage voltage, spot a puffed pack, explain disposal.

QUIZ

4 LESSONS, THEN THE QUIZ

1. LiPo basics
2. Battery math **DRILL**
3. Charging, storage and disposal
4. When a pack goes bad

10 questions · 80 % passes · no sign-off needed · retake any time

WHY LiPos are the biggest real fire risk in the hobby.

5

6

7

TIER 0 · PT. 6

Betaflight

Flash firmware, set motor direction and order, bind a remote.

BENCH

AT THE BENCH · 11 STEPS

Props OFF › Connect › Backup › Reset to defaults › Flash › Ports › Receiver › Modes › Motors › OSD › Check my work

Practised on the club Meteor 75 Pro at the kiosk, which checks the result against a stock config.

Photo DM'd to mtucker27@ · a mentor signs it off

WHY Most "won't fly" problems get solved here.

GATE 1

Meteor 75 Pro unlocked

TIER 1 · TINY WHOOP →

FLIGHT BUILD & TECH KNOWLEDGE & SAFETY CREW & LEADERSHIP

Quiz badges: pass at 80 %, no sign-off · Bench and witnessed badges: a mentor who holds the badge signs it off · demonstration, not attendance



FULL BADGE LIST

SCAN